

Writing proofs with sets

Definition: equality of sets

$$A = B \Leftrightarrow (A \subseteq B) \wedge (B \subseteq A)$$

$$A = \{x \in \mathbb{R} : x^2 < 5\}, B = \{x \in \mathbb{R} : x \leq 4\}$$

$$\text{let } x \in A: x^2 < 5$$

$$\Rightarrow -\sqrt{5} < x < \sqrt{5}$$

$$\text{But } (x < 5) \equiv (x \leq 4)$$

$$\therefore x \in B$$

$$\therefore A \subseteq B$$

Show how statements linked

$$x < \sqrt{5} < 4 \Rightarrow x < 4$$